The Cell and Its Organelles

Biology 101 - Lecture 1

The Cell Membrane

- The cell membrane is a phospholipid bilayer with embedded proteins.
- It is selectively permeable: it controls what enters and leaves the cell.

The Nucleus

- The nucleus contains the cell's DNA.
- The nucleolus inside the nucleus makes ribosomal RNA.

Making Proteins

- Ribosomes make proteins.
- The rough endoplasmic reticulum has ribosomes on its surface and makes proteins for export.
- The Golgi apparatus modifies, sorts and packages proteins.

Energy and Digestion

- Mitochondria produce most of the cell's ATP by aerobic respiration.
- Lysosomes contain digestive enzymes that break down worn-out organelles.
- The smooth endoplasmic reticulum makes lipids and detoxifies drugs.

Thank you